

# Lighting Models

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# Introduction

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## □ **Reflection model**

- Describes the interaction of light with a surface, in terms of the properties of the surface and the nature of the incident light
- Computes the light intensity at a object point. Works in world coordinate space
- Basic approach for a photorealistic presentation of a 3D object
- Local reflection model – considers only the interaction between a point on the surface being shaded and the light source
- Global reflection model – considers the indirect illumination (e.g. radiosity and ray tracing)

## □ **Illumination model**

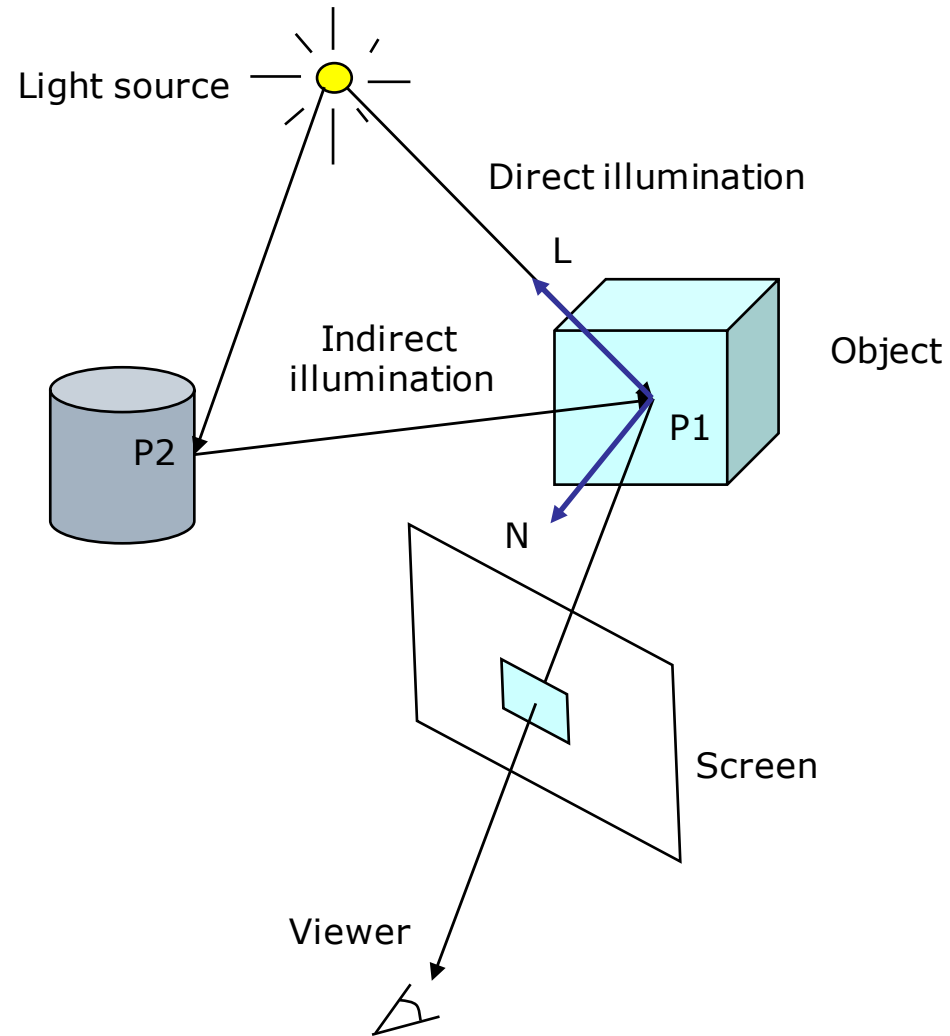
- Defines the nature of a light emanating from a source (e.g. color, geometry of the intensity distribution, etc)

## □ **Incremental shading technique**

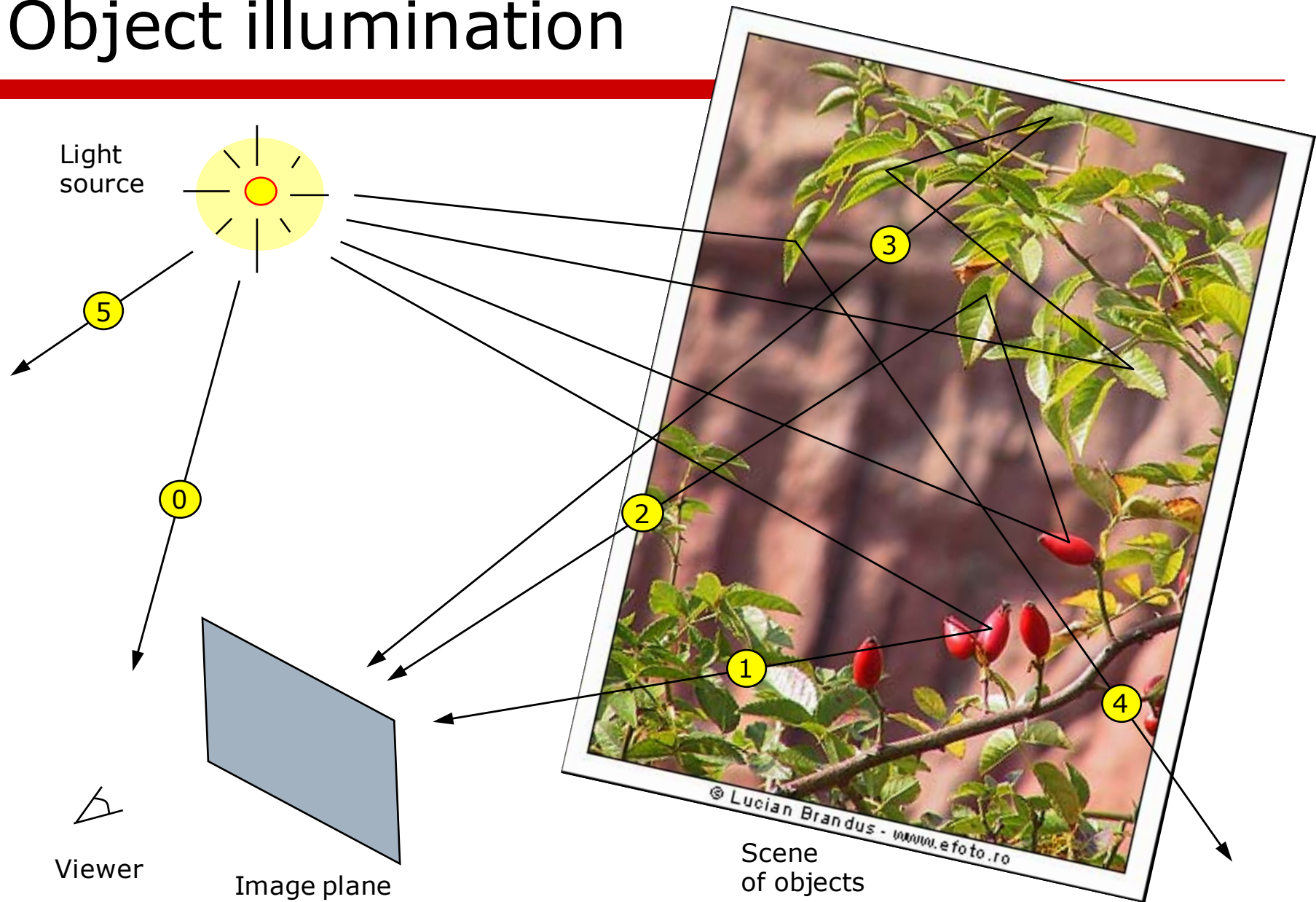
- Computes the light intensity in a pixel projection of a surface area
- Works in screen space

# Object illumination

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# Object illumination

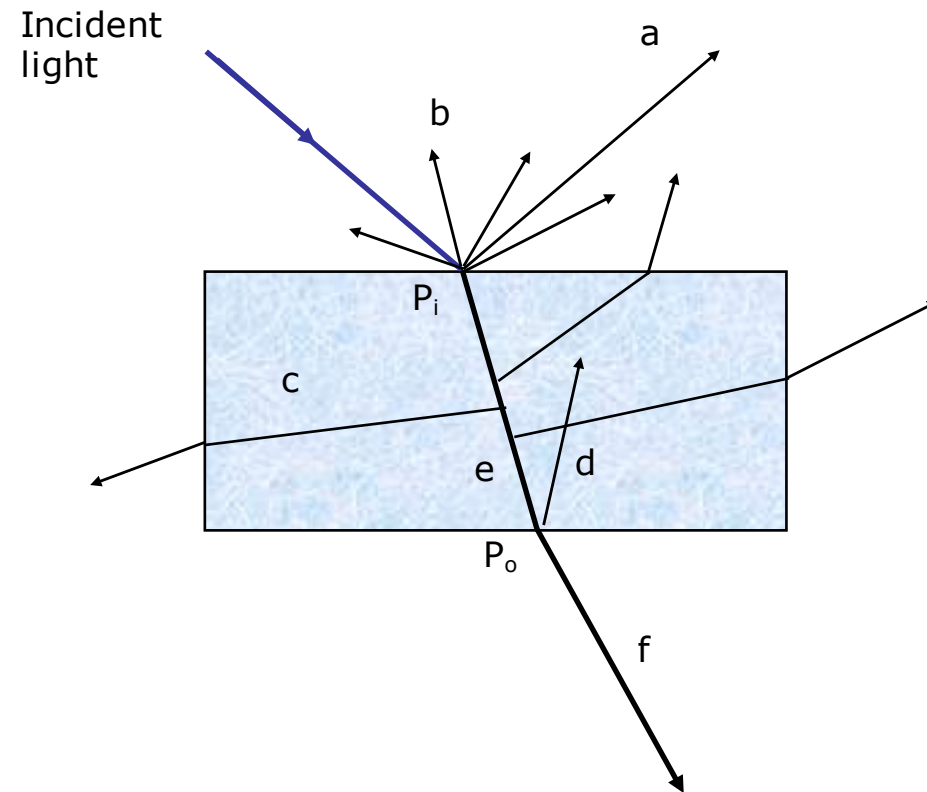


# Light interaction with a 3D object

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- a. Specular reflection
- b. Diffuse reflection
- c. Fluorescence emission
- d. Internal reflection
- e. Absorption
- f. Transmitted light (refraction)

Incident light =  
reflected light +  
scattered light +  
absorbed light +  
transmitted light



# Energy and light intensity

- Bidirectional Reflectivity Function (BDRF)
  - Describes qualitatively the reflection phenomenon
  - Assess the limitations of computer graphics models
  - Definition:  $R_{bd}(\lambda, \theta_i, \phi_i, \theta_v, \phi_v)$
  - $(\theta_i, \phi_i)$  – incoming light direction
  - $(\theta_v, \phi_v)$  – outgoing light direction toward the viewing direction.

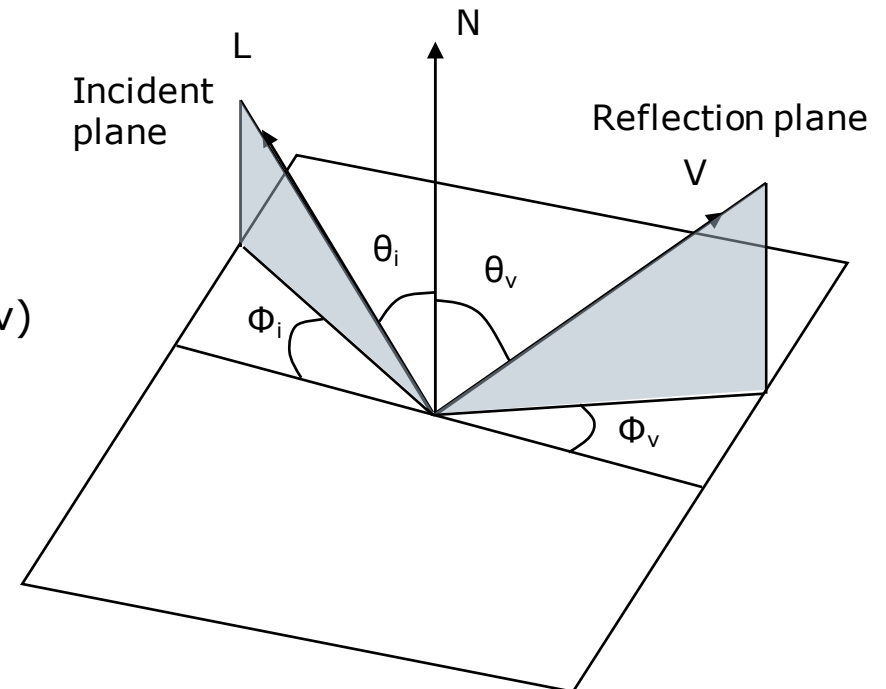
- Reflection model vs Radiation method:

$$I(\theta_v, \phi_v) = I_{\text{emitted}}(\theta_v, \phi_v) + I_{\text{reflected}}(\theta_v, \phi_v)$$

$$R_{bd}(\lambda, \theta_i, \phi_i, \theta_v, \phi_v) = \frac{I_v(\theta_i, \phi_i, \theta_v, \phi_v)}{E_i(\theta_i, \phi_i)}$$

$E_i(\theta_i, \phi_i)$  – incoming light energy

$I_v(\theta_i, \phi_i, \theta_v, \phi_v)$  – outgoing light intensity



# Energy and light intensity

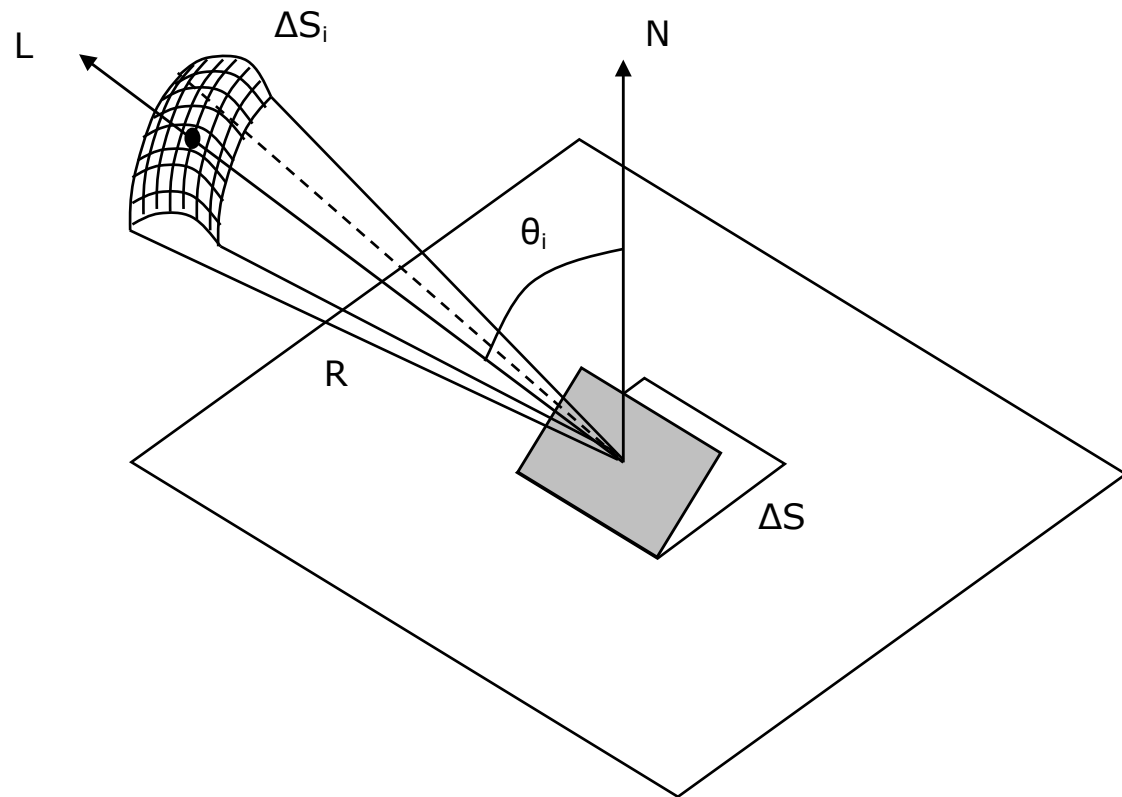
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- Incoming energy:

$$E_i(\theta_i, \Phi_i) = I_i(\theta_i, \Phi_i) d\omega_i \cos \theta_i$$

where  $\omega_i$  is the solid angle

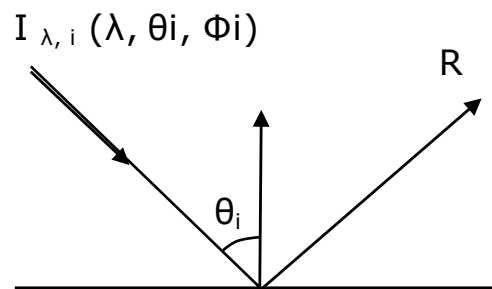
$$d\omega_i = \frac{\Delta S_i}{R^2}$$



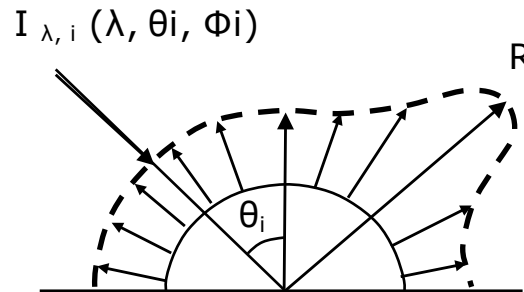


# Energy and light intensity

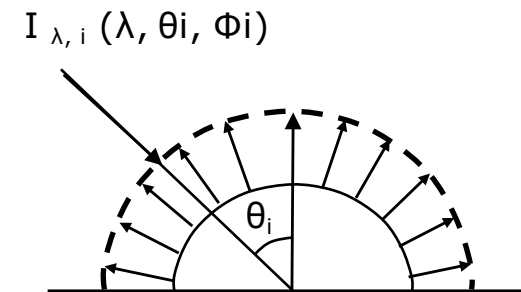
- BDRF is divided in three components:
  1. Specular contribution
  2. Directional diffuse contribution
  3. Ideal diffuse contribution
- Specular and directional diffuse components are due to first order reflections
- Ideal diffuse component is due to multiple reflections and subsurface intersections (microscopic variations in the surface)



(a) Specular reflection



(b) Directional diffuse reflection



(c) Ideal diffuse reflection

# Reflection models in graphics

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- Torrance and Sparrow (1967) introduced a surface model.
- Warnock (1967) computed the monochrome light intensity as a value attenuated by the distance between the light source and the object.
- Bouknight (1970) computed the light intensity as a function of the orientation of the polygonal surface against the light source. He introduced the diffusion Lambertian ( $\cos\theta$ ), to evaluate the constant light intensity of a polygonal area

$$I_{\lambda, r} = I_{\lambda, \text{ambient}} k_{\text{ambient}}(\lambda) + I_{\lambda, i} k_{\text{diffuse}}(\lambda) (N \cdot L)$$

- Phong reflection model, proposed by Phong Bui-Tuong in 1975
  - Considers just the first order reflection (i.e. direct illumination)
  - Has become a standard reflection model in the computer graphics community
- Blinn (1977) introduced an empirical specular term.
- Cook and Torrance (1982) concerned on accurate rendering of different metallic and non-metallic surfaces.

# Phong reflection model

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- Phong model = linear combination of three components: ambient, diffuse, and specular

# Ambient light source

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- Ambient component is an approximation to global illumination.
- In reality such surfaces would be illuminated by indirect illumination, such as multiple reflections from walls.
- In local reflection model the ambient component is modelled as a constant:

$$I_{\text{reflected}} = I_a k_a(\lambda).$$

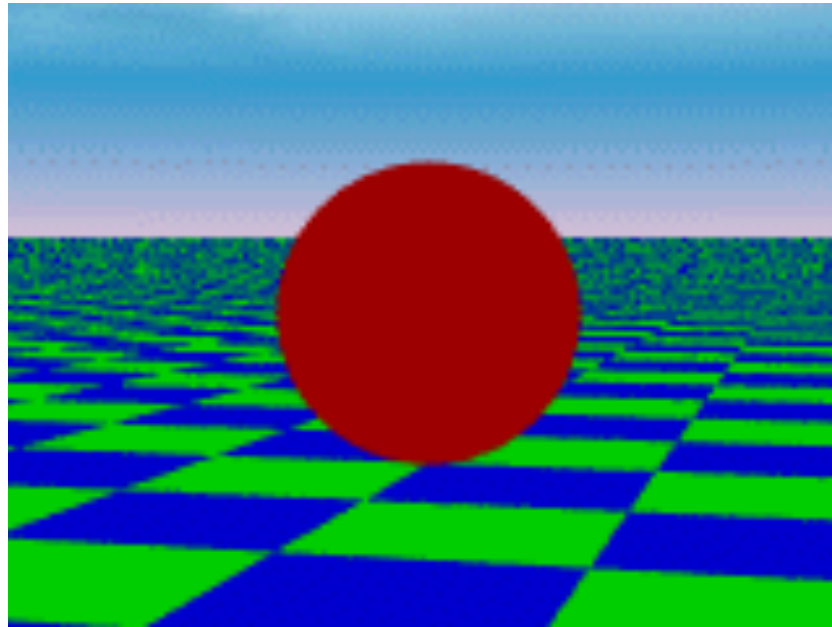
- Ambient Light source has no spatial or directional characteristics.
- Ambient light incident on each object is a constant for all the scene.
- Ambient light that is reflected by an object is independent of the object's position and orientation.
- Surface properties are used to determine how much ambient light is reflected. For RGB model they are  $k_{ar}$ ,  $k_{ag}$ ,  $k_{ab}$ .

Usually  $k_{\text{ambient}} = k_{\text{diffuse}}$

# Ambient light source

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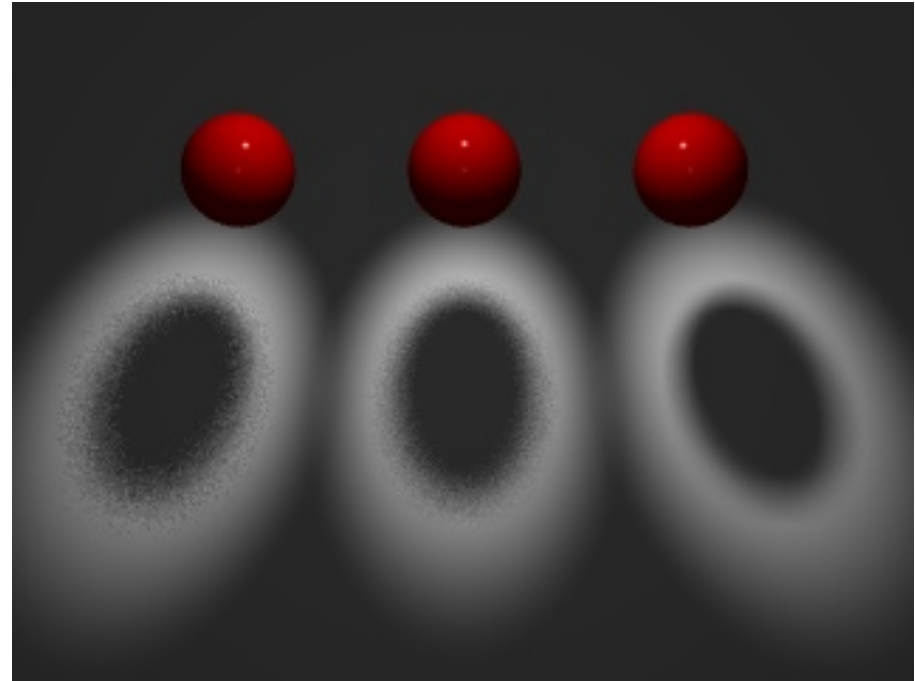
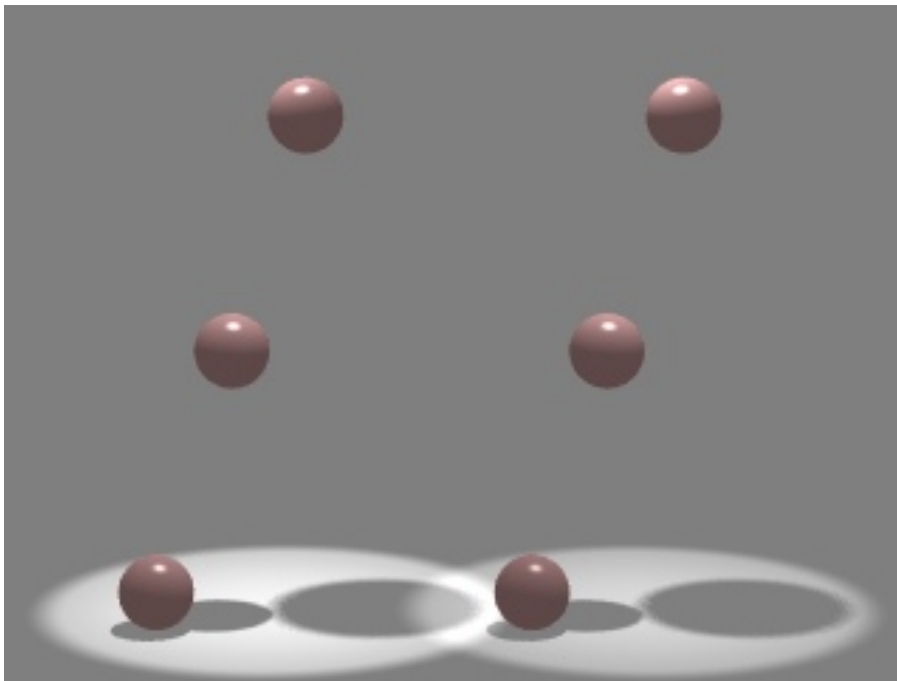
Examples from the course Computer Graphics, by Pascal Vuylsteker, AN Univ., 2003.



# Directional light source

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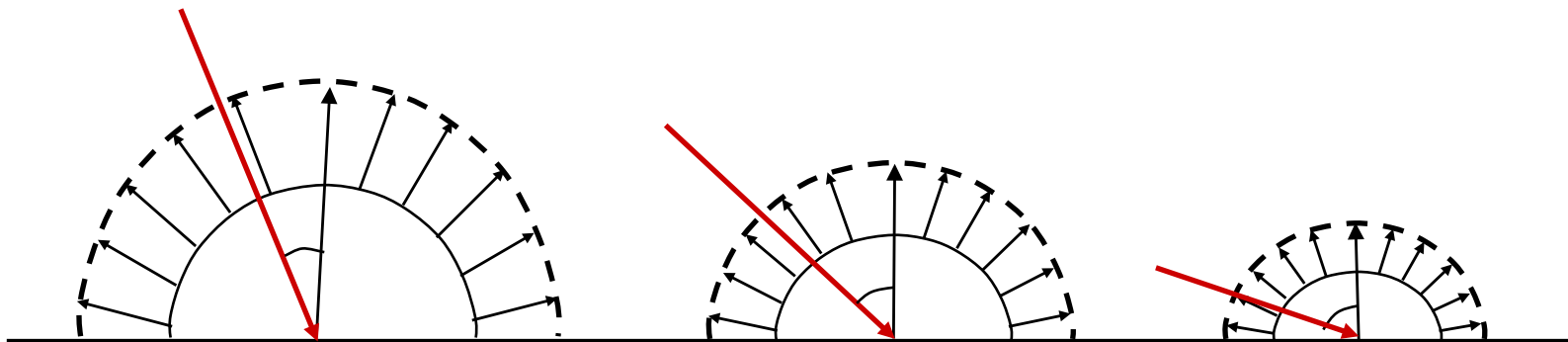
- Consider the direction of the light source to compute both the diffuse and specular components of illumination.
- With a directional light source the illumination direction is a constant.
- Point light source, spotlight, area light source.



# Ideal diffuse reflection

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- Lambert's Cosine law:
  - reflected energy from a small surface area in a particular direction is proportional to the cosine of the angle between that direction and the surface normal
- Ideal diffuse reflectors = Lambertian reflectors
  - Ideal diffuse reflectors reflect light according to Lambert's cosine law (Lambertian reflectors).
  - Energy that is reflected in any one direction is constant in Phong model. Reflected intensity is independent of the viewing direction.
  - Intensity depends on the light source's orientation relative to the surface.



# Diffuse component

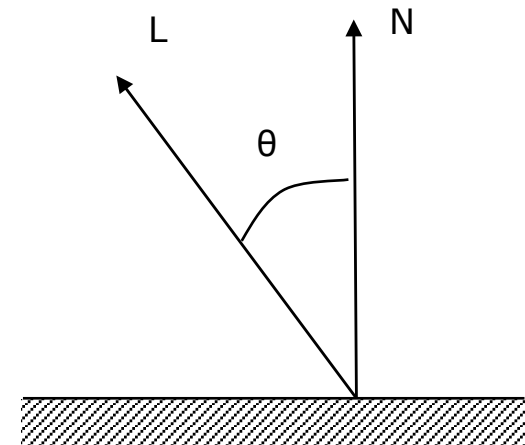
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$$I_{\text{diffuse}} = I_i (k_d(\lambda) \cos \theta)$$

$$I_{\text{diffuse}} = I_i (k_d(\lambda) (L \cdot N))$$

for m light sources:

$$I_{\text{diffuse}} = \sum_{j=1}^m I_{i,j} (k_d(\lambda) (L_j \cdot N))$$

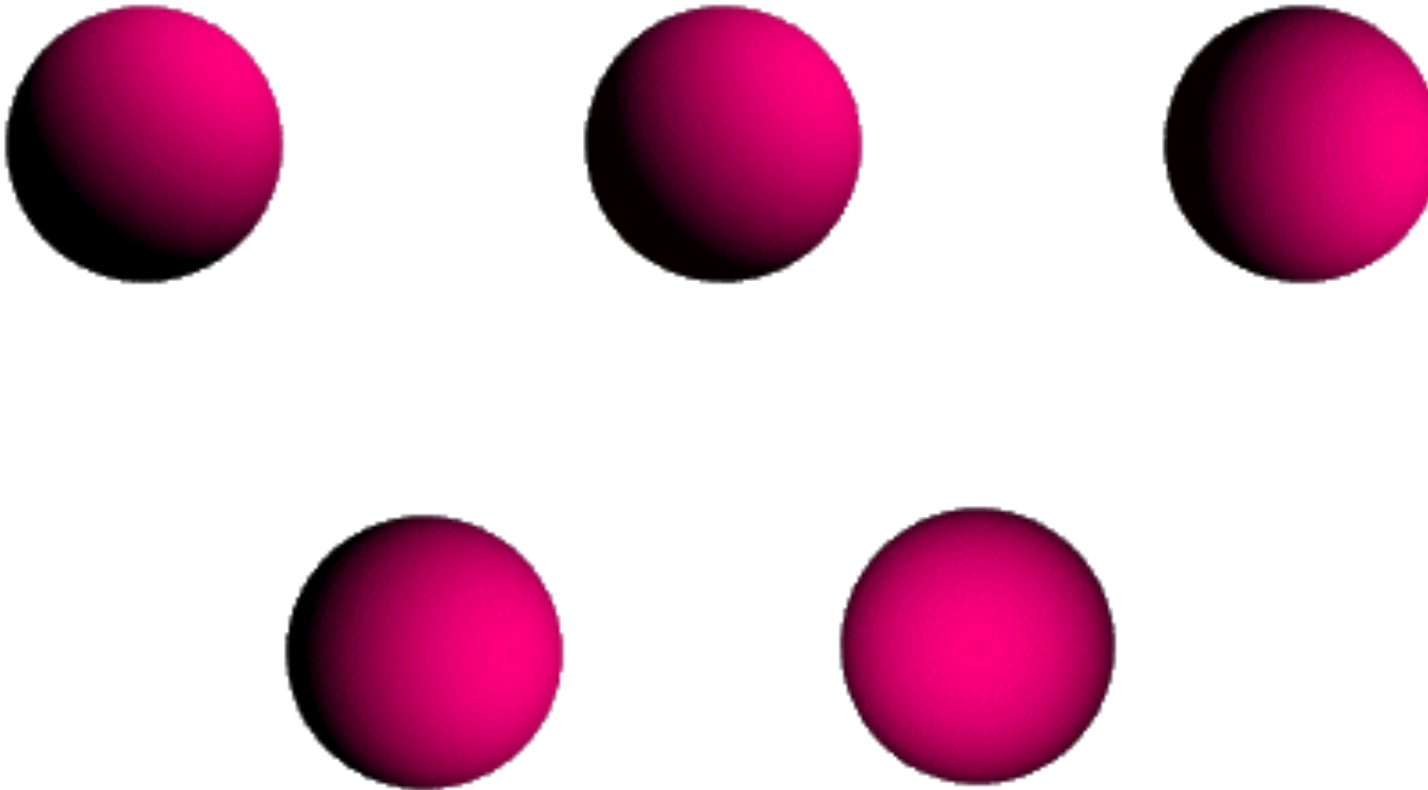




# Diffuse lighting examples

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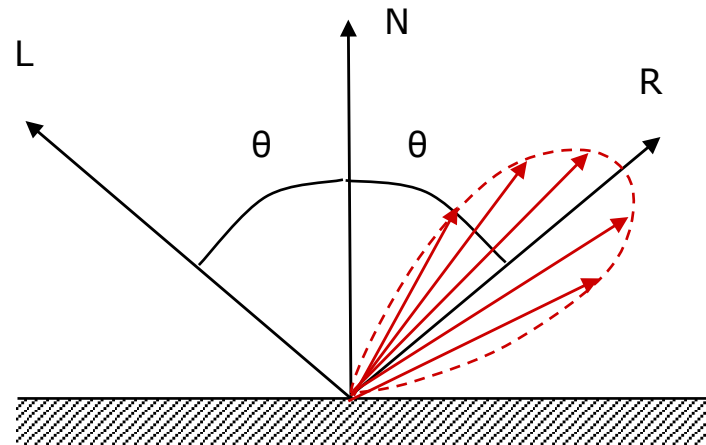
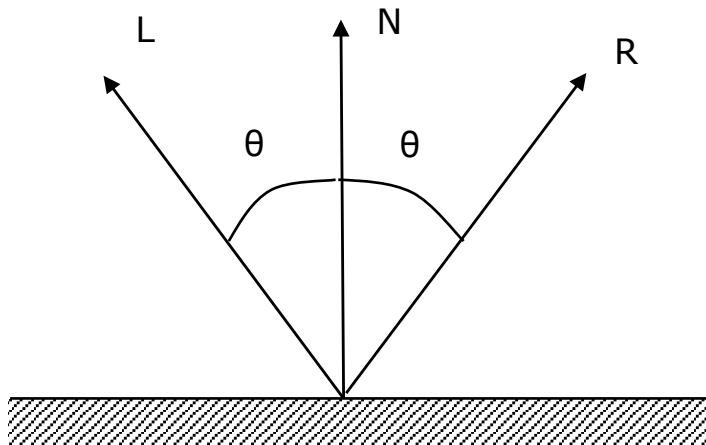
Examples from the course Computer Graphics, by Pascal Vuylsteker, AN Univ., 2003.



# Specular reflection

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- Specular reflector.
- Shiny surface, very smooth surface (e.g. polished metal).
- The specular reflector is an ideal mirror.



# Phong illumination model

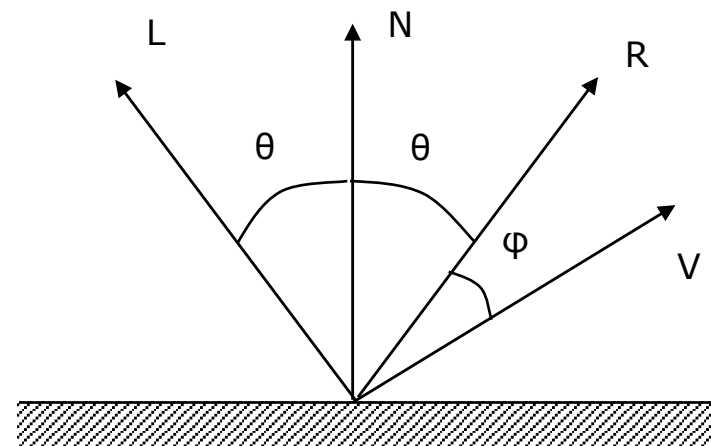
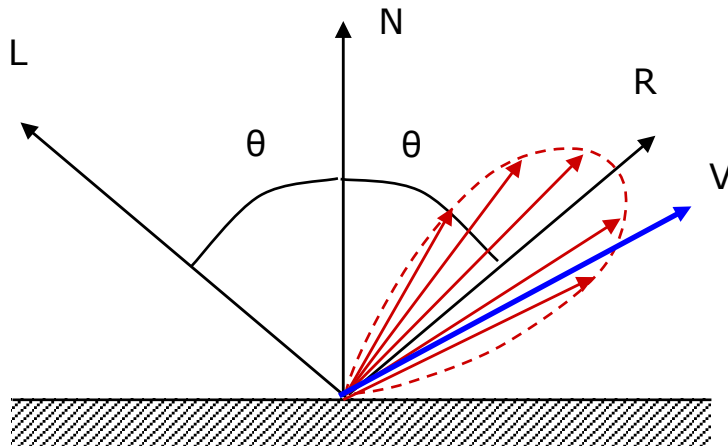
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- ❑ Specular reflection toward the viewing direction.
- ❑ No physical basis, but is the most used illumination model

$$I_{\text{specular}} = I_i k_s (\cos \phi)^n$$

$$I_{\text{specular}} = I_i k_s (R \cdot V)^n$$

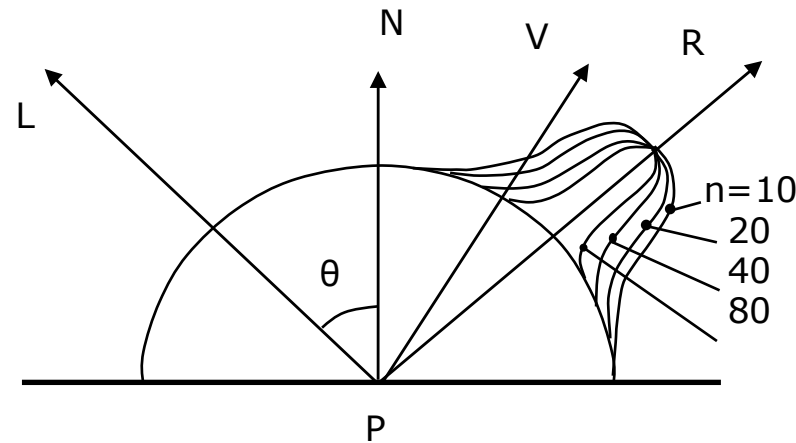
$n$  controls the surface shininess, the spread of the light rays around the  $R$  direction.



# Phong illumination model

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$n$  controls the surface shininess, the spread of the light rays around the  $R$  direction

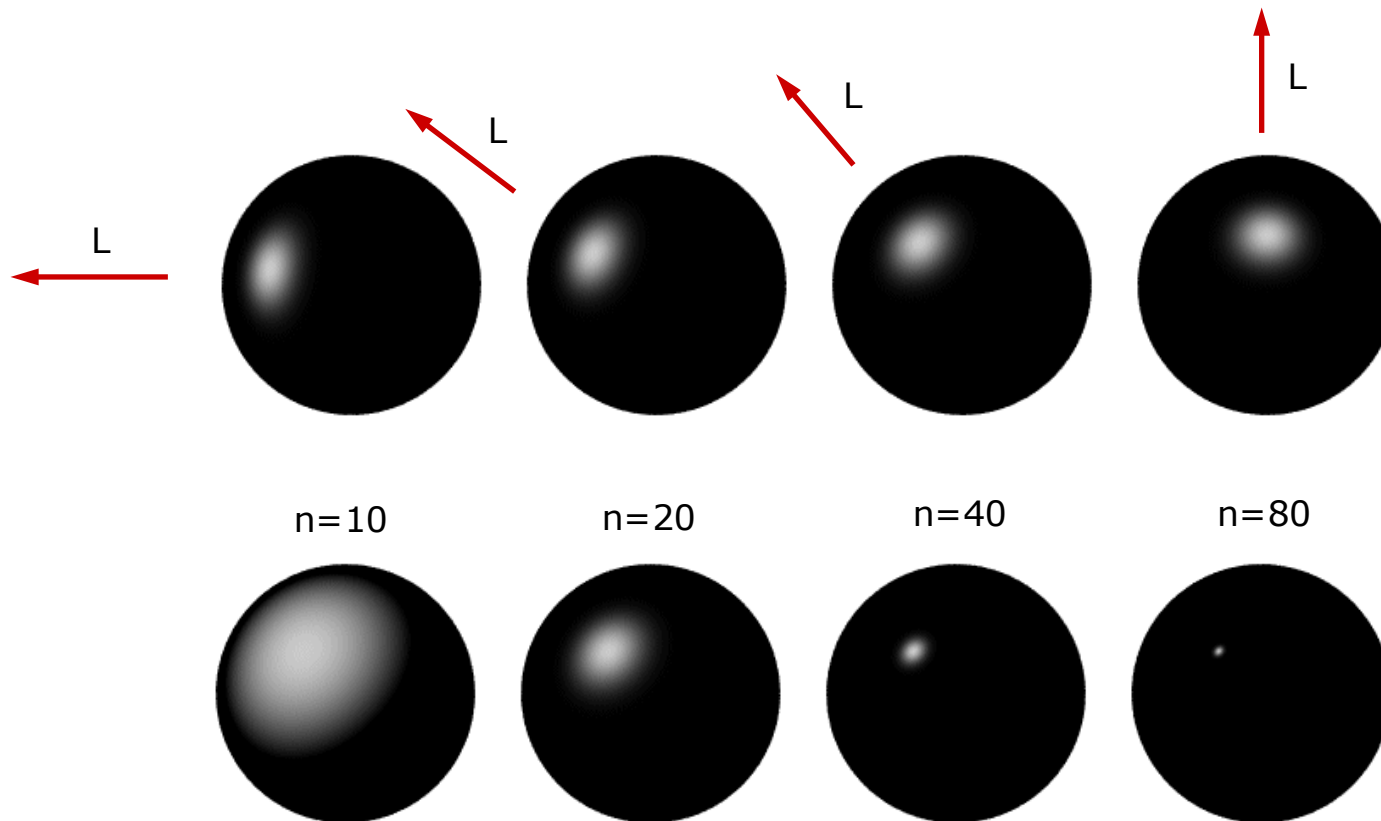


# Phong illumination examples

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Examples from the course Computer Graphics, by Pascal Vuylsteker, AN Univ., 2003.

- The following spheres illustrate specular reflections as the direction of the light source (L) and the coefficient of shininess (n) is changed



# Phong reflection model

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Consider all three components together: ambient diffuse and specular

$$I_v(\lambda) = I_a k_a(\lambda) + I_i k_d(\lambda) (L \cdot N) + I_i k_s (R \cdot V)^n$$

or as trigonometrical form:

$$I_v(\lambda) = I_a k_a(\lambda) + I_i k_d(\lambda) \cos \theta + I_i k_s (\cos \varphi)^n$$

for m light sources

$$I_v(\lambda) = I_a k_a(\lambda) + \sum_{j=1}^m I_{i,j} [k_d(\lambda) (L_j \cdot N) + k_s (R_j \cdot V)^n]$$

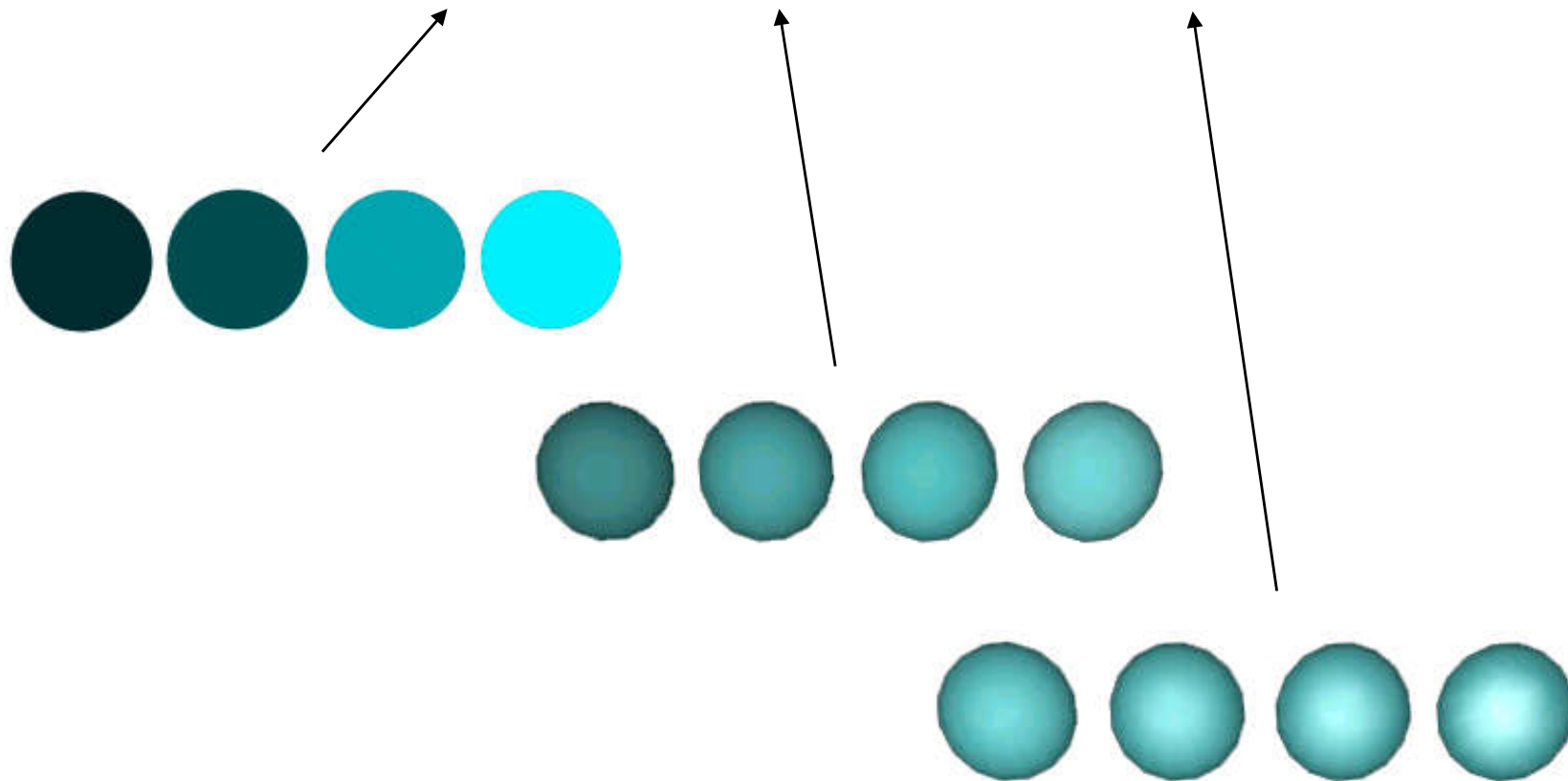
$$I_v(\lambda) = I_a k_a(\lambda) + \sum_{j=1}^m I_{i,j} [k_d(\lambda) (\cos \theta_j) + k_s (\cos \varphi_j)^n]$$

# Phong reflection model - examples

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Consider all three components together: ambient, diffuse and specular

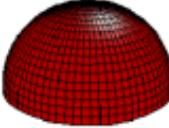
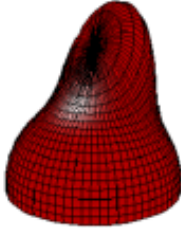
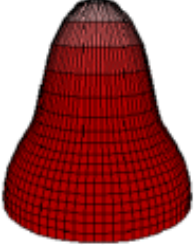
$$I_v(\lambda) = I_a k_a(\lambda) + I_i k_d(\lambda) (L \cdot N) + I_i k_s (R \cdot V)^n$$



# Phong reflection model - examples

Study taken from the course Computer Graphics, by Pascal Vuylsteker, AN Univ., 2003.

$$I_v(\lambda) = I_a k_a(\lambda) + I_i k_d(\lambda) \cos \Phi + I_i k_s (\cos \varphi)^n$$

| Phong               | $\rho_{\text{ambient}}$                                                             | $\rho_{\text{diffuse}}$                                                              | $\rho_{\text{specular}}$                                                              | $\rho_{\text{total}}$                                                                 |
|---------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| $\phi_i = 60^\circ$ |    |    |    |    |
| $\phi_i = 25^\circ$ |   |   |   |   |
| $\phi_i = 0^\circ$  |  |  |  |  |



# Phong model implementation

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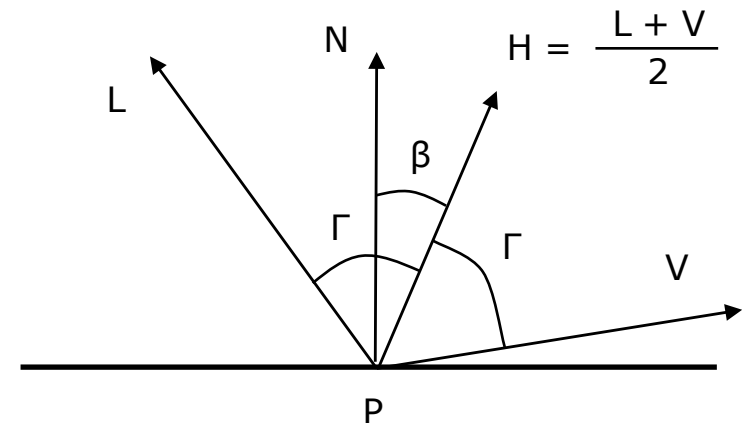
□ Assumptions for simplicity and computation time:

1. Light source and the view point are at infinity, then the L and V vectors are constant over the domain of the scene
2. Introduce a vector H, and the dot product ( $N \cdot H$ ) instead of ( $R \cdot V$ ). H is the unit normal vector to a hypothetical surface that is oriented in a direction halfway between the light direction vector L and the viewing vector V

$$H = (L+V)/2$$

The angle between R and V is  
twice the angle between N and H:

$$\varphi = 2\beta$$



# Phong model implementation

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Consider all three components together: ambient diffuse and specular

$$I(\lambda) = I_a k_a(\lambda) + I_i k_d(\lambda) (L \cdot N) + I_i k_s (N \cdot H)^n$$

or as trigonometrical form:

$$I(\lambda) = I_a k_a(\lambda) + I_i k_d(\lambda) \cos \theta + I_i k_s (\cos \beta)^n$$

for m light sources

$$I(\lambda) = I_a k_a(\lambda) + \sum_{j=1}^m I_{i,j} [ k_d(\lambda) (L_j \cdot N) + k_s (N \cdot H_j)^n ]$$

$$I(\lambda) = I_a k_a(\lambda) + \sum_{j=1}^m I_{i,j} [ k_d(\lambda) (\cos \theta_j) + k_s (\cos \beta_j)^n ]$$

# Colors in Phong model

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- Color as light intensity is a function of wavelength

$$I(\lambda) = I_a k_a(\lambda) + I_i k_d(\lambda) (L \cdot N) + I_i k_s (N \cdot H)^n$$

- In the RGB system the color is:  $I = \alpha_r I_r + \alpha_g I_g + \alpha_b I_b$
- Consider the specular highlights as white
- Control the color of the objects by appropriate setting of the diffuse coefficients, and the scene by the ambient coefficients
- It should be computed three intensity equations:

$$I_r = I_a k_{ar} + I_i [k_{dr} (L \cdot N) + k_s (N \cdot H)^n]$$

$$I_g = I_a k_{ag} + I_i [k_{dg} (L \cdot N) + k_s (N \cdot H)^n]$$

$$I_b = I_a k_{ab} + I_i [k_{db} (L \cdot N) + k_s (N \cdot H)^n]$$

Conclusion:

$$I(r,g,b) = I_a k_a(r,g,b) + I_i [k_d(r,g,b)(L \cdot N) + k_s (N \cdot H)^n]$$

# Conclusions on Phong model

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- Phong local reflection model:

$$I(\lambda) = I_a k_a(\lambda) + I_i k_d(\lambda) (L \cdot N) + I_i k_s (N \cdot H)^n$$

or 
$$I(\lambda) = I_a k_a(\lambda) + I_i k_d(\lambda) \cos \theta + I_i k_s (\cos \beta)^n$$

- Light sources are point sources
- Light sources and the viewer are located at infinity
- The diffuse and specular terms are modeled as local components
- An empirical model (shininess  $n$ ) is used to simulate the decrease of the specular term around the reflection vector modeling the glossiness of the surface
- The color of the specular reflection is assumed to be that of the light source. It means the highlights are rendered white regardless of the material
- The global term that is the ambient contribution is a constant

# Questions and problems

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1. What is the surface model for the object illumination?
2. How the Phong model describes the multiple reflections of the light rays?
3. What is the location of the light source for the ambient component?
4. What means the Lambertian surface?
5. Why the diffuse illumination infers the light source position?  
How the light source position is described within the Phong model?
6. What is the difference between R and V directions of the specular reflection in the Phong model?
7. Why the specular illumination infers the light source position?  
How the light source position is described within the Phong model?

# Questions and problems

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8. Why the specular light is white and the ambient and diffuse components are colored?
9. How is computed the illumination by three light sources by Phong model, if one is red, one is green, and another is blue?
10. How is computed the illumination by three light sources by Phong model, if one is red, one is magenta, yellow, and another is cyan?